Panayiota Papafiggou

PhD Researcher – Intelligent Diagnostics & Condition Monitoring
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Research Profile

PhD researcher in Intelligent Diagnostics and Condition Monitoring at the University of Sheffield. My research focuses on AI-based diagnostics for fleet-based condition monitoring of electric vehicle motors, with broader interests in artificial intelligence, machine learning, reinforcement learning, multi-agent systems, and autonomous intelligent systems. Previous research spans multi-agent reinforcement learning, computational modelling, large-scale simulation, maritime data analysis, and formal verification.

Education

University of Sheffield **Starting Oct 2026**
PhD in Electrical and Electronic Engineering | EPSRC Studentship *Sheffield, UK*

- Research: *AI-Based Diagnostics for Fleet-Based Condition Monitoring of Electric Vehicle Motors Using Machine Learning Frameworks.*
- Supervisors: Dr Panagiotis Panagiotou and Dr Georgios Tsialiamanis.

University of Warwick **Sep 2025 – Sep 2026**
MSc Computer Science *Coventry, UK*

- Dissertation: *Assortative Matching and the Emergence of Cooperation in Multi-Agent Reinforcement Learning Systems.*
- Relevant study: Machine Learning, Data Analytics, Agent-Based Systems, Algorithmic Game Theory, and Artificial Intelligence.

University of Nottingham **Sep 2022 – Jun 2025**
BSc (Hons) Computer Science *Nottingham, UK*

- Dissertation: *A Study of Synthetic Differential Geometry in Type Theory.*
- Relevant study: Artificial Intelligence, Software Engineering, Computer Security, Human–AI Interaction, Databases, and Formal Methods.

Research Experience

MSc Dissertation Researcher **Sep 2025 – Sep 2026**
University of Warwick *Coventry, UK*

- Investigated how assortative matching and partner-selection mechanisms influence cooperation among reinforcement learning agents in repeated Prisoner's Dilemma environments.
- Developed baseline and hybrid multi-agent models in Python using Q-learning, adaptive partner switching, and assortative rematching.
- Designed and conducted large-scale simulation experiments including sensitivity analyses, population studies, parameter sweeps, ablation studies, convergence analysis, and behavioural evaluation.
- Analysed cooperation, learned strategies, stability, switching behaviour, and the effects of interaction structure on emergent multi-agent behaviour.

External Research Collaborator **Jun 2025 – Aug 2025**
Cyprus University of Technology *Limassol, Cyprus*

- Developed and tested algorithms using Automatic Identification System (AIS) data to detect vessel movement in maritime areas of interest.
- Processed noisy, incomplete, and irregular maritime observations, considering reporting errors, missing observations, irregular transmissions, and message constraints.
- Conducted research under the supervision of Associate Professor Herodotos Herodotou.

Undergraduate Dissertation Researcher **Sep 2024 – Jun 2025**
University of Nottingham *Nottingham, UK*

- Investigated Synthetic Differential Geometry using Homotopy Type Theory, dependent type theory, and the Agda proof assistant.
- Developed mechanically verified definitions and proofs involving curvature, metrics, transformations, and geometric structures.
- Produced formal implementations and visualisations supporting computer-assisted mathematical proof development.

Industry Experience

Software Engineer Intern **Jun 2024 – Jul 2024**
Amdocs *Limassol, Cyprus*

- Supported software testing and automation using Unix, Ginger by Amdocs, Octane, Jenkins, and Postman.
- Worked with CI/CD automation, API testing, debugging, software verification, and validation.

Summer Intern **Jun 2023 – Sep 2023**
Golden Union Shipping *Limassol, Cyprus*

- Supported migration to a cloud-based crew-management platform through business-process analysis and documentation.
- Verified migrated data and supported system-transition and regulatory-compliance activities.

Selected Technical Projects

Multi-Agent Coordination in Space Exploration
Developed a three-dimensional rover simulation in Python using Ursina and compared PPO, MADDPG, and evolutionary algorithms under terrain, energy, navigation, and coordination constraints.

NLP-Based Travel Booking Chatbot
Built a conversational AI system in Python incorporating intent recognition, information retrieval, sentiment analysis, and natural-language interaction for travel-booking scenarios.

Technical Skills

Programming	Python, Java, C, C++, SQL, JavaScript
AI & Machine Learning	Machine Learning, Reinforcement Learning, Q-Learning, Multi-Agent Systems, PPO, MADDPG, Evolutionary Algorithms, Natural Language Processing
Diagnostics & Engineering	Intelligent Diagnostics, Condition Monitoring, Data-Driven Fault Detection, AI for Engineering Systems
Research Methods	Computational Modelling, Experimental Design, Sensitivity Analysis, Ablation Studies, Statistical Analysis, Behavioural Analysis, Large-Scale Simulation
Scientific Computing	NumPy, pandas, Numba, Matplotlib, PettingZoo, Ursina
Formal Methods	Agda, Homotopy Type Theory, Dependent Type Theory, Formal Proofs, Mathematical Logic
Software & Tools	Git, GitHub, Linux, Unix, Jenkins, CI/CD, Postman, MySQL, Agile Development

Leadership & Activities

Computing Officer	MedTech Society, University of Nottingham; supported technical operations for interdisciplinary events and workshops.
MedTech Hackathon	Co-organised an interdisciplinary hackathon involving students from computing, medicine, engineering, and related disciplines.
European Youth Parliament	Participated in public speaking, collaborative policy discussions, teamwork, and debate.
Academic Societies	Warwick AI Society, Warwick Computing Society, Coding Society, HackSoc, and Greek Cypriot Society.
Sport & Music	AEL Volleyball Team, Cyprus; Classical Guitar, Grade 5.

Languages

Greek: Native **English:** Fluent